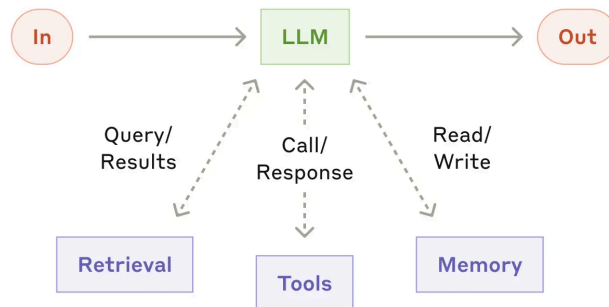


Tutorial Week 2

Limitations of traditional LLM

- Inability to perceive the external environment
- Inability to effectively handle complex tasks (Interlocking tasks)
- Knowledge base is lagging

Agent can be a good solution to these problems!



Agent operation mode

1. ReAct (Reasoning and Acting)

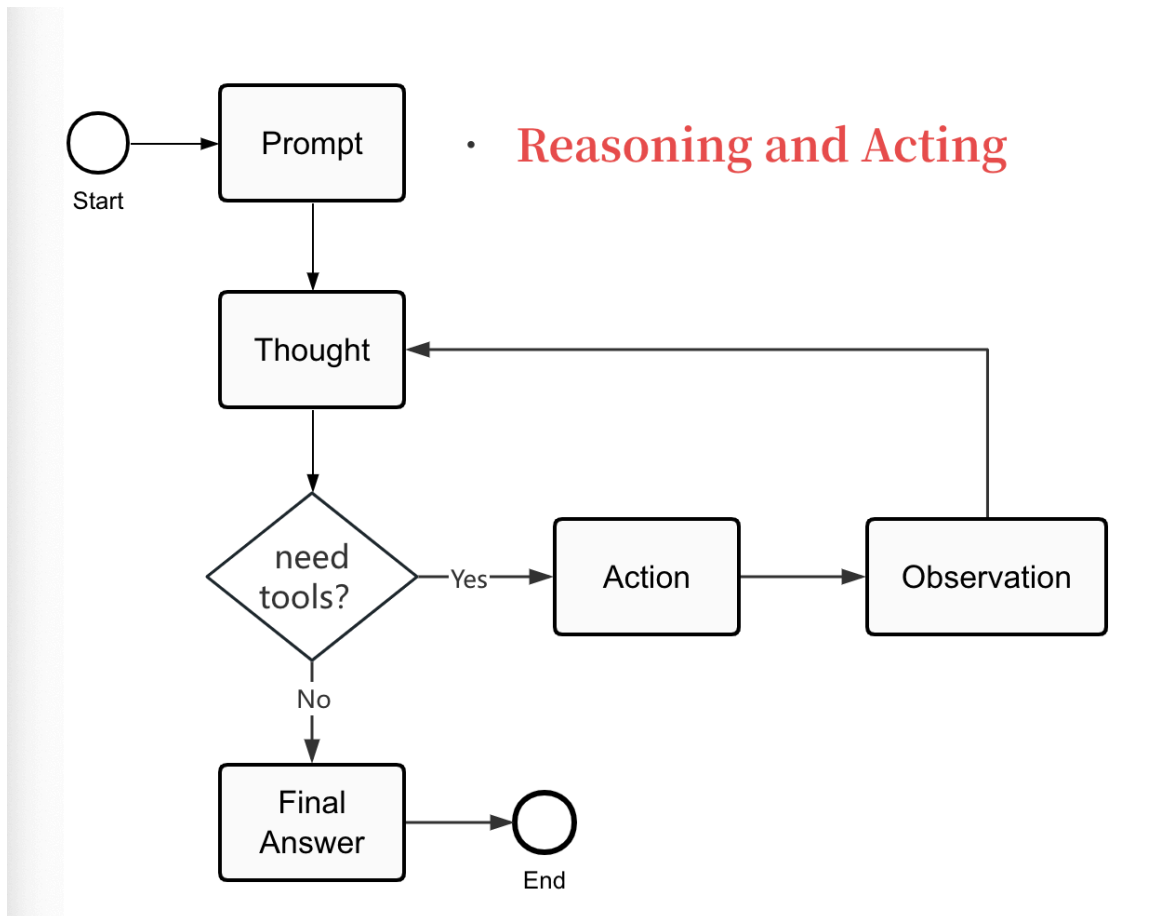
This framework was proposed by Princeton & Google Brain team, published in ICLR 2023

ReACT: Synergizing Reasoning and Acting in Language Models

This framework aims to enable Large Language Models (LLMs) to perform both "Reasoning" and "Acting" when accomplishing tasks, in order to compensate for the shortcomings of the traditional reasoning-only or action-only approaches. By alternately generating thought chains and task behaviors, the ReAct paradigm

achieves higher accuracy and stronger generalization capabilities across multiple tasks, while improving interpretability and controllability.

1.1 ReAct Architecture



1.2 How to design Prompt to let LLM learn how to call tools?

XML (Extensible Markup Language)

XML is a markup language used to standardize the storage and transfer of data, is highly extensible (you can define your own tags), and is easy to parse.

```
<?xml version="1.0" encoding="UTF-8"?>
<students>
  <student id="1">
    <name>Alice</name>
    <age>20</age>
```

```
</student>
<student id="2">
  <name>Bob</name>
  <age>22</age>
</student>
</students>
```

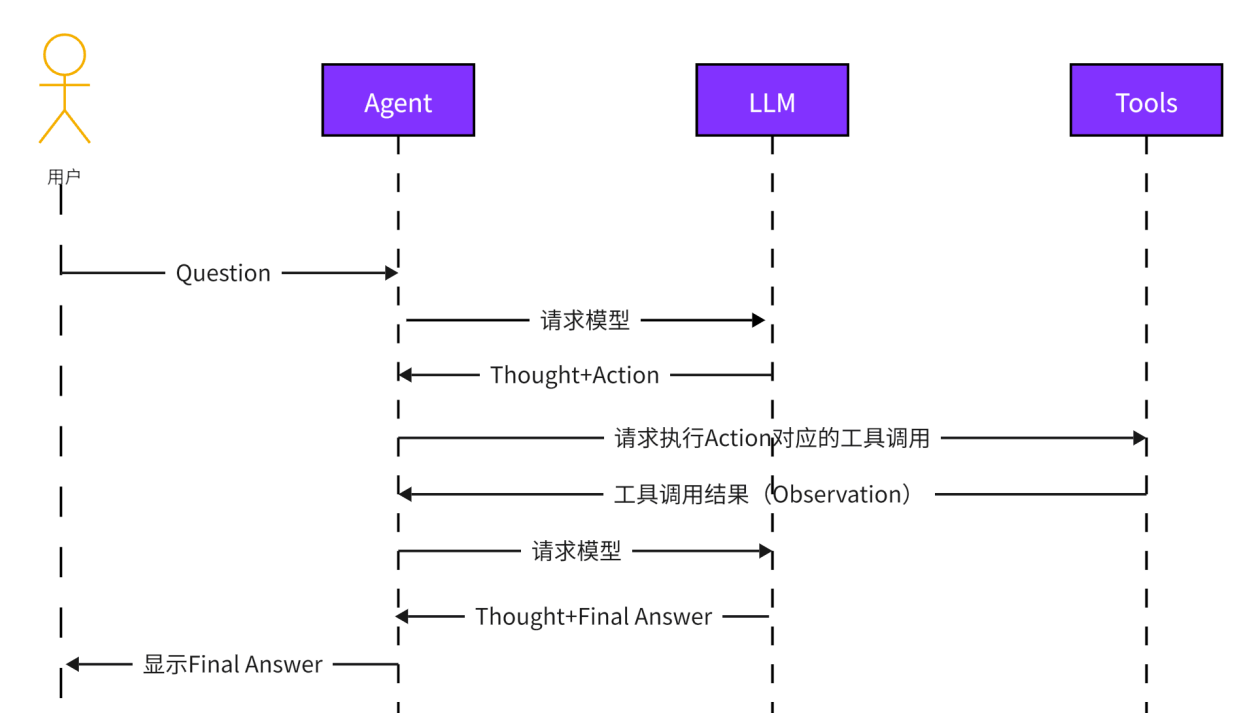
In the ReAct schema, we mainly use five kinds of custom xml tags, used to constrain the data transfer format of the Agent

- <question> 用户问题
- <thought> 思考
- <action> 采取的工具操作
- <observation> 工具或环境返回的结果
- <final_answer> 最终答案

Example illustration (How tall is the Eiffel Tower?)

```
<question>埃菲尔铁塔有多高？</question>
<thought>我需要找到埃菲尔铁塔的高度。可以使用搜索工具。</thought>
<action>get_height("埃菲尔铁塔")</action>
<observation>埃菲尔铁塔的高度约为330米（包含天线）。</observation>
<thought>搜索结果显示了高度。我已经得到答案了。</thought>
<final_answer>埃菲尔铁塔的高度约为330米。</final_answer>
```

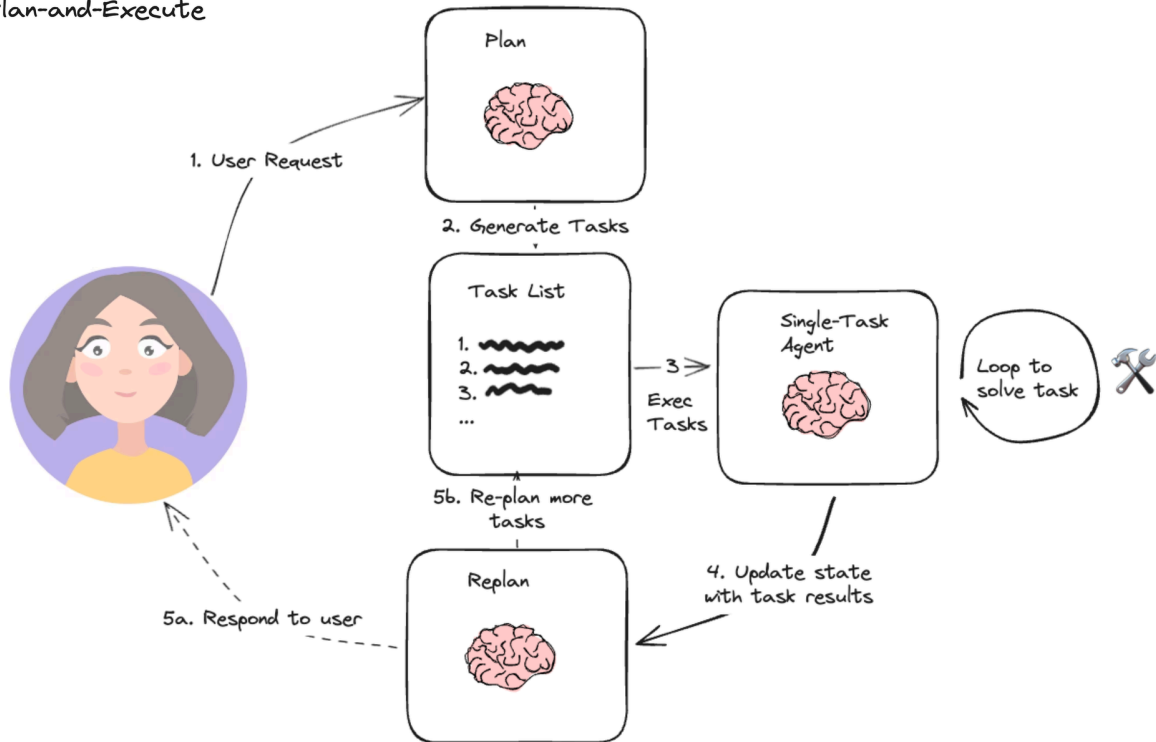
1.3 ReAct Runtime Sequence Diagram



2. Plan-And-Execute

Plan-and-Execute is to decouple planning and execution to avoid LLM losing context or going off track in long tasks.

Plan-and-Execute



2.1 A simple case

- **Plan**

- The Agent first generates a complete **Plan**, usually a sequence of subtasks, based on the user's requirements.
- For example:

User Q: Help me write an article about "AI in Education".

Plan:

1. Collect the latest research on AI in education.
2. Summarize common application scenarios
3. Discuss advantages and disadvantages
4. Generate a draft of the article

- **Execute.**

- The Agent executes subtasks incrementally according to a schedule.
- Each subtask may call a tool (search API, database, calculator, etc.) or it may call LLM again.
- The results can be logged during execution and the output can be consolidated at the end.

2.2 Advantages of Plan-and-Execute

- **Structured:** Compared to ReAct, you won't get "lost" in long conversations, and it is more suitable for complex tasks.
- **Controllability:** You can review or modify the plan and hand it over to the Agent for execution.
- **Scalable:** Agents with different roles can be responsible for different parts (e.g., one Planner Agent, one Executor Agent).
- **Reduce Token Consumption:** You can plan once and pass only the subtask context when executing multiple steps.

2.3 Comparison between ReAct and Plan-and-Execute

Characteristics	ReAct	Plan-and-Execute
Thinking style	Think + Act alternately at each step	Plan and execute
Suitable Scenarios	Short tasks, exploratory tasks	Long tasks, multi-step tasks, structured tasks
Contextual stress	High (LLM has to look at full history each time)	Low (execution only has to look at localized tasks)
Controllability	Low	High (plans can be reviewed/modified)

Practice: First Agent Application (ReAct)

//Clone the repository

```
git clone https://github.com/kiritoAlex/2025-Fall-CS309-Lab.git
```

//Create a new conda environment

```
conda create -n cs309 python=3.10
```

//Enter conda environment

```
conda activate cs309
```

//Install necessary packages

```
pip install openai tiktoken python-dotenv requests click
```

```
cd week2
```

```
python agent.py snake
```